

Technical Communication

Investigation of compressibility of rockfill materials using DEM: Role of sample size and boundary condition

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ARTICLE INFO

Keywords:

Rockfill materials
Discrete element method
Size effect
Compressibility
Boundary condition

ABSTRACT

Understanding the in-situ behavior of rockfill materials through laboratory tests is challenging due to the influence of sample size. In this study, the discrete element method (DEM) is utilized to investigate the effects of sample size and boundary condition on the compressibility of rockfill materials at both macroscopic and microscopic scales. The results reveal that rockfill compressibility increases with sample size when rigid boundaries are applied, but no significant size effect is observed for periodic boundaries. Besides, the one-dimensional compression behavior of different initial packing varies with sample size under rigid boundaries, with the variance decreasing as size increases; however, this effect is negligible under periodic boundaries. Additionally, both the distribution uniformity of contact number and fabric anisotropy increase with increasing sample size under rigid boundary conditions. At a microscopic level, it can be observed that the sample size effect of granular materials is correlated to the coordination number per unit volume $CN/(1 + e)$ for the considered particle shapes.

1. Introduction

Rockfill materials are indispensable in dam construction. However, the dimension of rock fragments often exceeds the permissible limits of experimental apparatuses, posing challenges in understanding their in-situ mechanical behavior through traditional laboratory techniques (Indraratna et al., 1993; Varadarajan et al., 2003; Fu et al., 2018; Ning et al., 2020). Parsons (1936) conducted pioneering work on sample size effects, which shows that the friction angle of sand decreases as the size of the direct shear box increases. In the following, numerous experiments have been conducted to investigate the sample size effect (Holtz et al., 1956; Varadarajan et al., 2003; Stark et al., 2018; Shen et al., 2021). However, no consensus has been reached regarding the minimal sample size. Table 1 presents the recommended maximum particle size d_{max} used in various tests from different standards. The discrepancies among the standards highlight the need for further research into the sample size effect.

Conducting mechanical experiments on granular materials often involves the consideration of multiple factors, and it is critical to isolate individual variables and study their effects while maintaining others

controllable. The discrete element method (DEM) proves to be a numerical tool for exploring the mechanical behavior of granular materials in a manageable manner (Yu and Shen, 2012; Liu et al., 2019; Oskooei et al., 2021). Various studies have been carried out to understand the influence of sample size on the compressibility of rockfill materials. Kuhn and Bagi (2009) showed sample's peak compressive strength decreases with increasing sample size for both rigid and periodic boundaries, while sample's stiffness decreases with increasing sample size only for rigid boundary. Additionally, Gu et al. (2017) suggested the ratio of specimen size to the mean particle size to be larger than 11.5 to avoid the potential scale effect on small strain modulus. Wang et al. (2022) suggested the recommended specimen diameter to be larger than 6 times the largest particle size for determining the parameters of Duncan and Chang's E-B constitutive model. He et al. (2023) revealed that the width of shear band is dependent of sample size and the size ratio should be no less than 25 for investigating strain localization. Cantor and Ovalle (2023) found that the size ratio should be no less than 15 to obtain stable critical state shear strength and void ratio which is larger than 10 as recommended by ASTM D6528. However, most previous studies focused on narrow particle size distributions with d_{max}/d_{min} smaller than 5. It

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Table 1

Recommended maximum particle size $d_{\max}$ used in various tests from different standards (modified after Cantor and Ovalle, 2023).

Standard name	Test name	Recommended maximum $d_{\max}$
ASTM D3080	Direct shear	Min ($H/6$, $D/10$)
ASTM D7181	Triaxial test	$D/6$
ASTM D6528	Simple shear	$H/10$
ASTM D6467	Ring shear	$H/10$
BS 1377	Direct shear	$H/10$
BS 1377	Triaxial	$D/5$
BS 1377	Ring shear	$H/5$ and $T/25$
JGS 0561	Direct shear	$H/23.5$ (The size can be reduced for soils with a wide particle size distribution.)
JGS 0525	Triaxial	$D/20$ for $D = 35$ to 100 mm
JGS 0530	Triaxial	$D/10$ for $D = 300$ mm
JGS 0525/JGS 0530	Triaxial	$D/5$ for soils having a wide range of particle sizes
GB/T 50123-2019	Triaxial	$D/10$ for $D \leq 61.8$ mm; $D/5$ for $D \geq 101$ mm
GB/T 50123-2019	Direct shear	$D/12 \sim D/8$ & $H/8 \sim H/4$ for $d_{\max} \leq 60$ mm

Note that D , H , and T are the sample diameter, height, and width of the ring, respectively.

motivates the research that considers broader particle size distributions and larger sample sizes that closely mimic experimental setups despite the significant increase in particle numbers and the associated computational cost.

Given the work reviewed above, this study adopted a three-dimensional discrete element model to investigate the influence of sample size and boundary condition on the compressibility of rockfill materials. Various sample sizes, particle shapes, boundary conditions, and densities are considered for simulations during which the evolution of the distributions of contact and fabric are tracked. A correlation between the coordination number and the macroscopic compression modulus is established for the considered sample sizes and particle shapes.

2. DEM model

This study adopted the rolling resistance linear contact model which is an efficient alternative to emulate the influence of particle shape in DEM simulations containing more than 10,000 particles (Iwashita and Oda, 1998; Zhou et al., 1999; Gu et al., 2020; Tang and Tao, 2022; Wu

et al., 2022). Rigid and periodic boundaries were both used in one-dimensional compression tests. Fig. 1 (a) shows that the sample with rigid boundary is a cylinder of equal height H and diameter L imitating the physical sample in the experiment, while the sample with periodic boundary is cubic with side length L . Fig. 1 (b) shows the particle size distribution of Lianghekou rockfill materials and the numerical samples for DEM simulations in this work. Then $L/d_{\max}$ denotes the ratio of specimen diameter L to maximum particle size $d_{\max}$. Since the experimental gradation curve excludes particles with diameters less than 5

Table 2

Numerical parameters adopted in the DEM simulations.

Model parameters	Value
Particle density	2760 kg/m ³
Wall stiffness	1×10^{10} N/m
Particle normal stiffness k_n	$k_n = k_0 \times r$
Particle shear stiffness k_s	$k_s = \alpha \times k_n$
Wall-particle friction coefficient	0.0
Interparticle friction coefficient μ	0.45
Rolling resistance coefficient μ_r	0.50

Note: $k_0 = 8 \times 10^8$ N/m², $\alpha = 0.15$, and r is the radius of the particle.

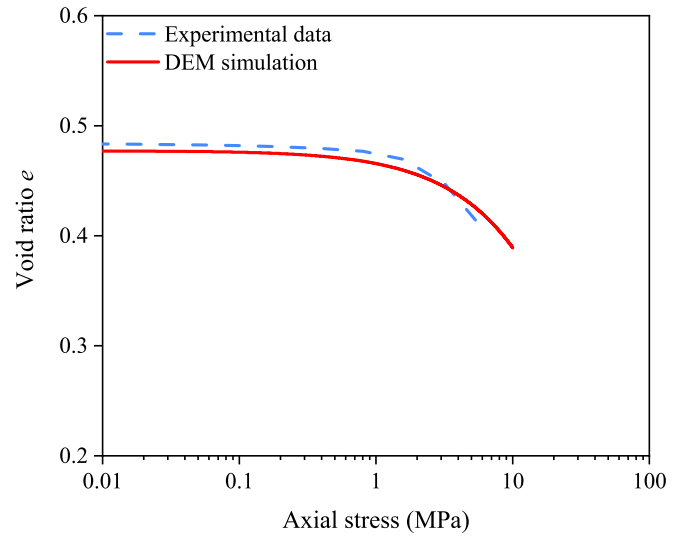


Fig. 2. One-dimensional compression curves from experiment and DEM simulation.

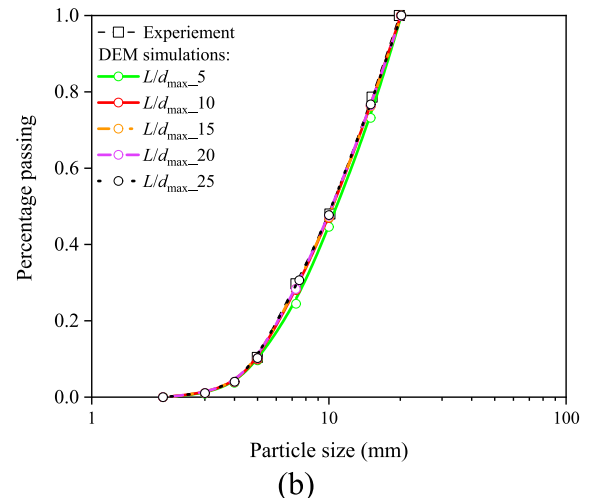
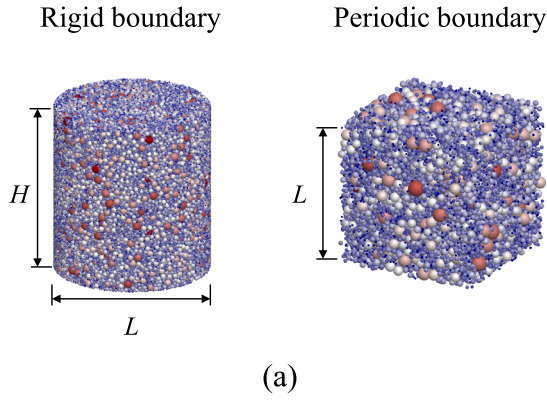


Fig. 1. The setup of the discrete element model: (a) sample configurations and (b) particle size distributions.

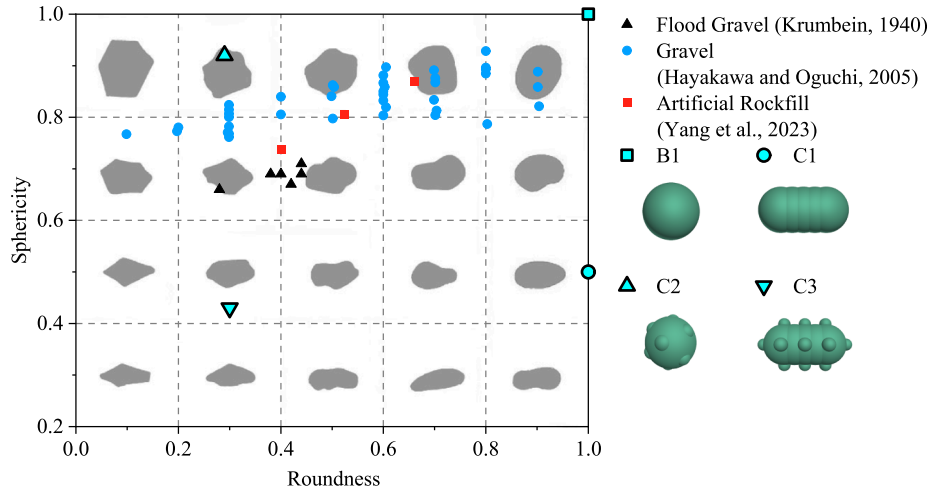


Fig. 3. Distribution of shape indices of the selected particles in the DEM simulations.

Table 3
Simulation program in this study.

Particle label	Particle type	$L/d_{\max}$	void ratio e	Boundary condition type
B1	Sphere	5	0.577, 0.540	rigid
		10	0.540, 0.510, 0.479	rigid
		15	0.537, 0.489, 0.482	rigid
		20	0.535, 0.482	rigid
		25	0.535, 0.477	rigid
		5	0.544	periodic
C1	Clump consisting of 5 equal spheres	10	0.542	periodic
		15	0.540	periodic
		5	0.481	rigid
C2	Clump consisting of a central sphere of radius r_1 and 12 surrounding spheres of radius r_2 ($r_2 = 0.286 r_1$)	10	0.444	rigid
		5	0.575	rigid
C3	Clump consisting of 5 central sphere of radius r_1 and 14 surrounding spheres of radius r_2 ($r_2 = 0.3 r_1$)	10	0.530	rigid
		5	0.670	rigid
		10	0.583	rigid

mm, this study takes the minimum particle size as 2 mm. Additionally, the $d_{\max}/d_{\min}$ of 10 is larger than those in previous studies (Huang et al., 2014; Gu et al., 2017; Shao et al., 2020). The radius expansion method was adopted and the detailed procedure can be found in Hu et al. (2023). The number of particles ranges from 2,116 to 263,537 in the simulations, which is relatively higher than those reported in existing literature.

A strain rate of 0.1/s was applied to the top wall to compress the specimen, with the bottom wall fixed, which is lower than the loading rate recommended in previous studies (Cil and Alshibli, 2014; Das and Das, 2019). During the one-dimensional compression, the inertia number I of the specimen was ensured to be less than 10^{-3} for attaining a quasi-steady state (da Cruz et al., 2005; Lopera Perez, J. C. et al., 2016).

To enhance comparability with experimental data for DEM parameter calibration, the same sample size with a similar void ratio and rigid boundary was generated in DEM simulations. Table 2 shows the DEM model parameters adopted in this paper. Fig. 2 compares the one-dimensional compression behaviors obtained from the experiment and the DEM simulation. The result shows that the adopted contact parameters reflect the compression behavior of rockfill materials sampled from the Lianghekou hydropower station project. Particle fragmentation was not considered in this study. Additionally, various particle shapes were selected for the further validation of conclusions drawn based on the simulation of spherical particles. Fig. 3 shows four representative particles with shape indices; the shape indices of gravel and artificial

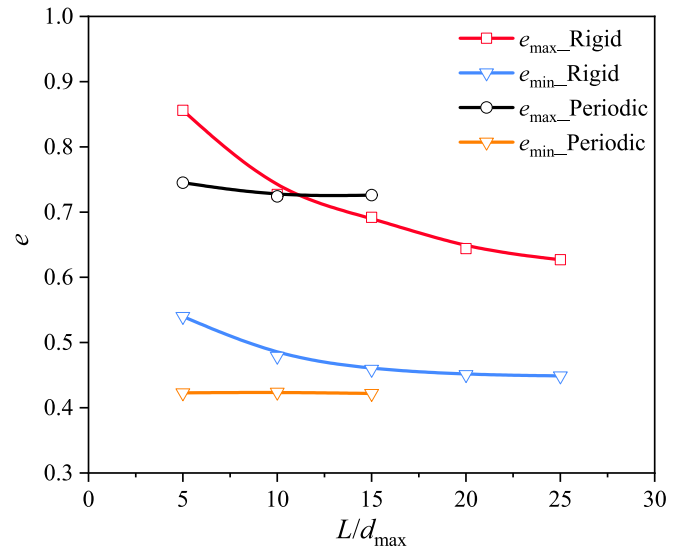


Fig. 4. The variation of maximum and minimum void ratios with sample size using rigid and periodic boundaries for B1 samples.

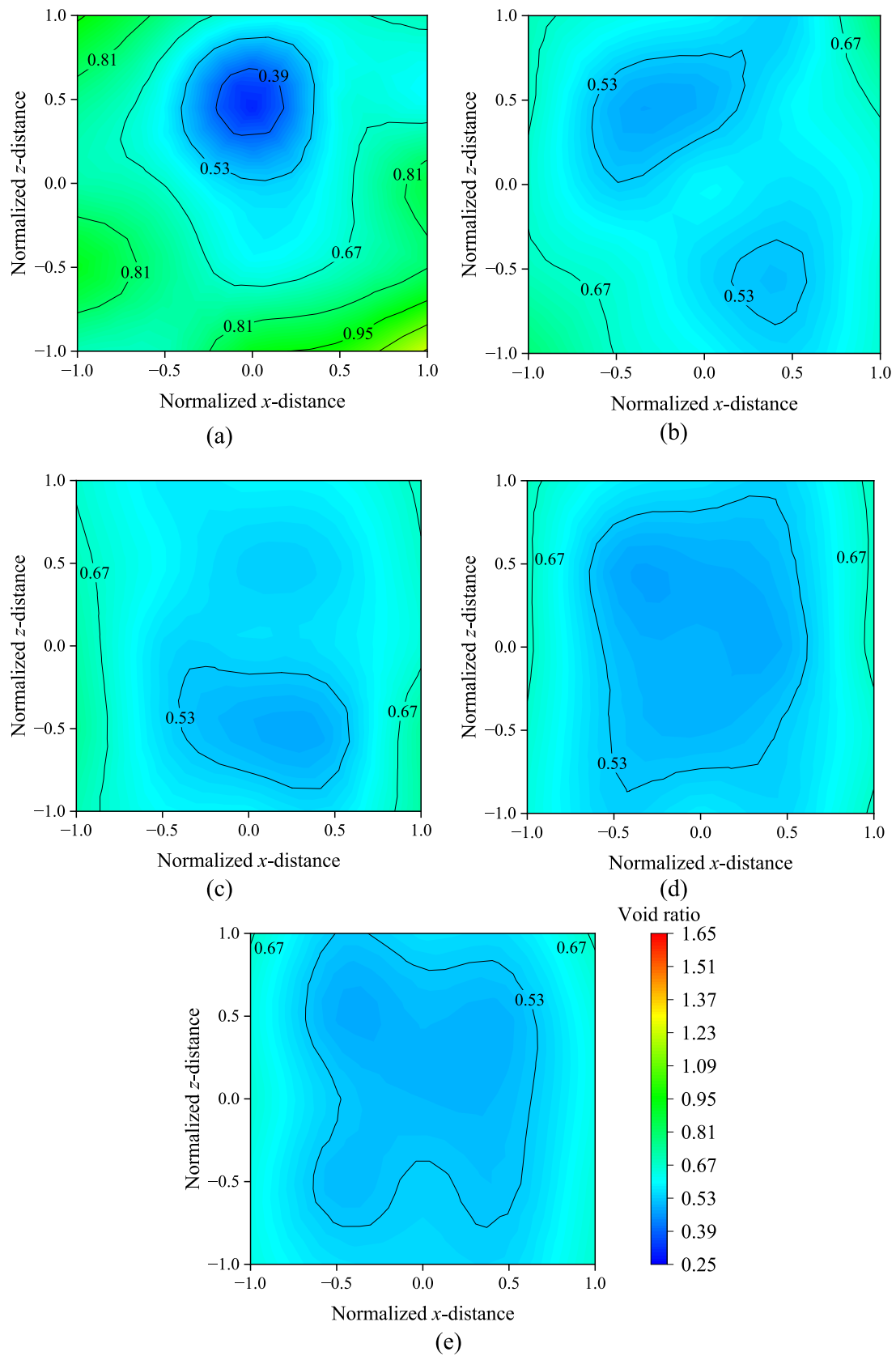


Fig. 5. The void ratio distribution at the central cross-section of the sample at its loosest state using rigid boundary: $L/d_{\max}$ equals (a) 5; (b) 10; (c) 15; (d) 20 and (e) 25 for B1 samples.

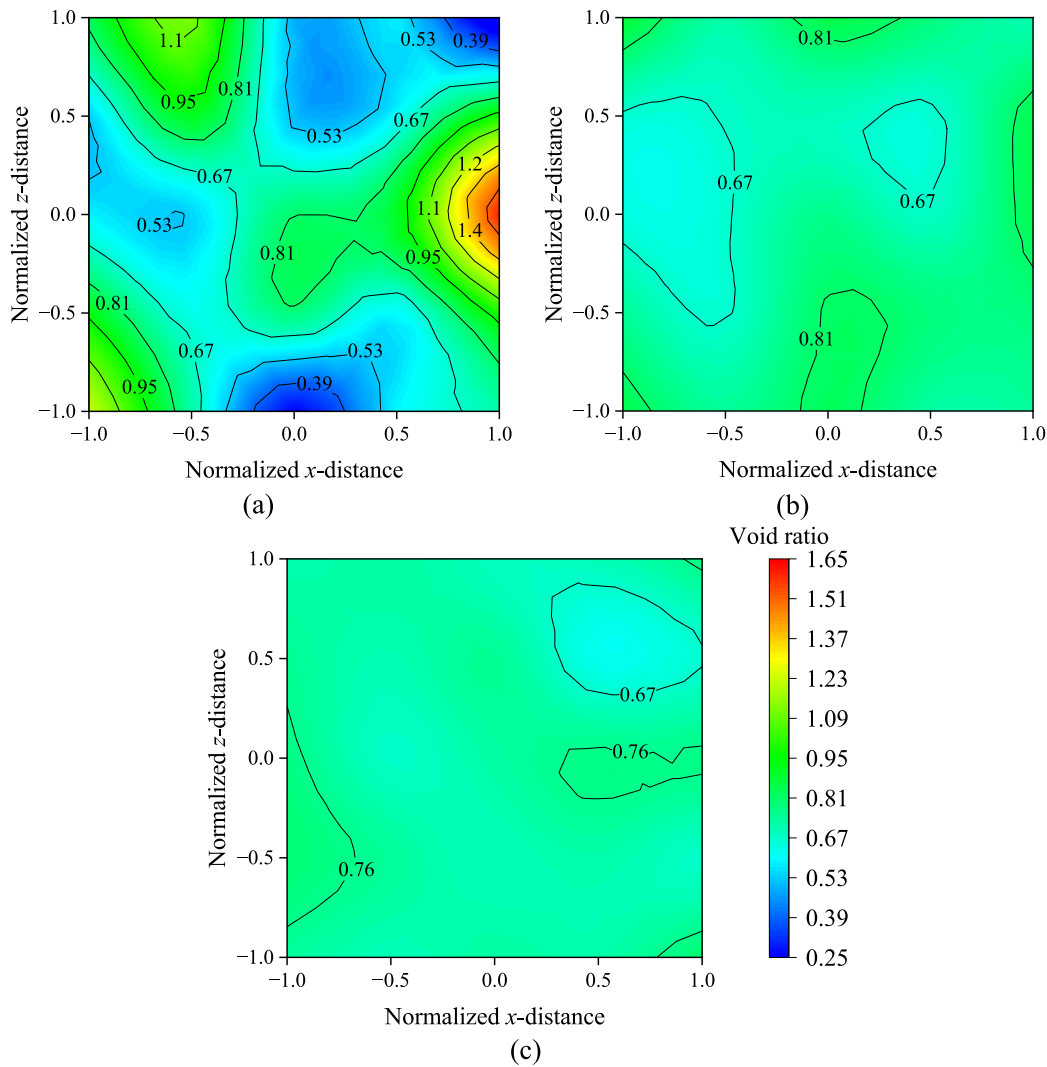


Fig. 6. The void ratio distribution at the central cross-section of sample at its loosest state using periodic boundary: $L/d_{\max}$ equals (a) 5; (b) 10; (c) 15 for B1 samples.

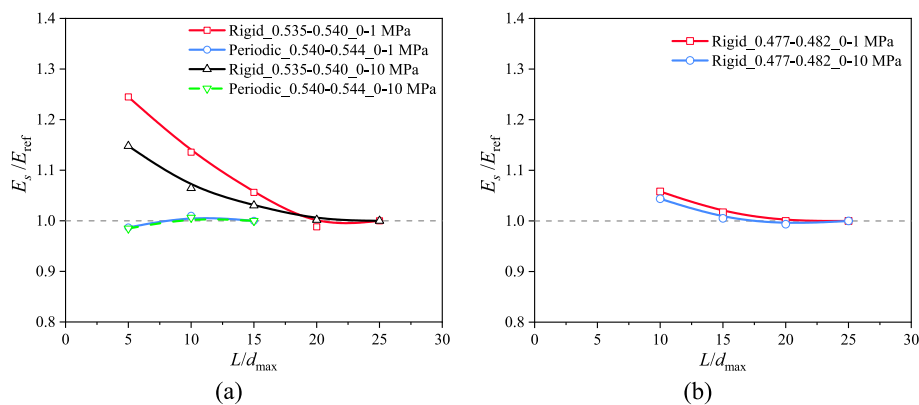


Fig. 7. Variation of secant compression modulus ratio with different $L/d_{\max}$ at void ratios ranging from: (a) 0.535–0.540 and (b) 0.477–0.482 for B1 samples.

rockfill are also shown for comparison (Krumbein, 1940; Hayakawa and Oguchi, 2005; Yang et al., 2023). Moreover, the rockfill sampled from the Dashixia rockfill dam shows sphericity ranging from 0.70 to 0.97 (An et al., 2022). The comparison indicates that the selected particle shape could represent the particle shapes of rockfill materials. The designed simulation program is summarized in Table 3. In the designed simulation program, two void ratio ranges, i.e. 0.535 to 0.540 and 0.477 to

0.482, were chosen for comparing the compressibility for different sample sizes. Besides, representative samples with relative densities from 84 % to 88 % were prepared to investigate the difference between the efficacy of two indices, i.e. void ratio and relative density, for gauging sample size effect. The representative relative density is close to the field condition for dam construction reported in previous works (Varadarajan et al., 2003; Honkanadavar and Sharma, 2014; Gupta,

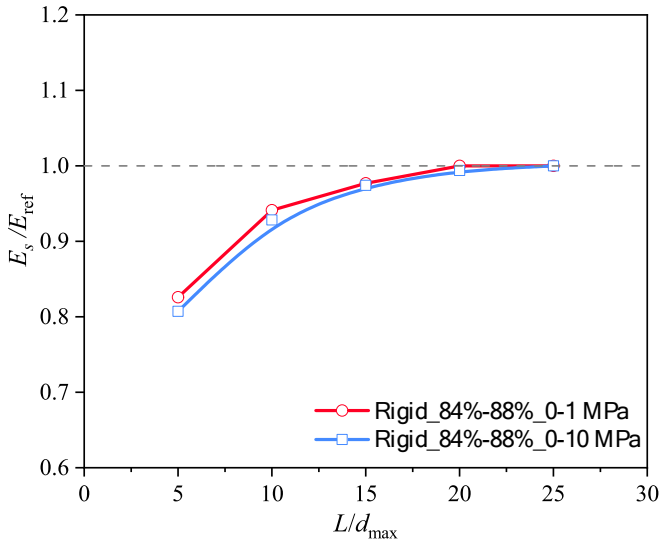


Fig. 8. Variation of secant compression modulus ratio against L/d_{max} for B1 particle samples with relative densities ranging from 84% to 88%.

2016). The samples consisting of irregular particles, i.e. clumps, were generated to explore the potential linear relation between microscopic parameters and compression modulus obtained using spherical particles.

3. Results and discussions

3.1. Macroscopic analysis

Fig. 4 shows the variation of maximum and minimum pre-compression void ratios with sample size with rigid or periodic boundaries for samples of spherical particles, i.e. B1 particles. The maximum void ratio e_{max} and minimum void ratio e_{min} refer to those obtained before the one-dimensional compression test. The maximum void ratio was obtained under a confining pressure of 5 kPa using an interparticle friction coefficient of 0.45 and a rolling resistance coefficient of 0.5, while the minimum void ratio was obtained under the same pressure using an interparticle friction coefficient and a rolling resistance coefficient both of zero (Gu et al., 2020). For rigid boundary, the maximum and minimum void ratios decrease as the sample size increases. In contrast, there is no notable change in maximum and minimum void ratios with varying sample sizes for periodic boundary condition. The minimum void ratio for the sample with rigid boundary tends to converge to 0.45, and it is close to the minimum void ratio e_{min} of 0.42 of the sample with periodic boundary. The maximum void ratios for the two boundary conditions decrease as the sample size increases, while

the maximum void ratio under rigid boundary is smaller than that under periodic condition as L/d_{max} increases. Figs. 5 and 6 show the void ratio distributions at the central cross-section of B1 particle samples of different sample sizes at their loosest state for the rigid boundary and periodic boundary conditions, respectively. It is shown that the rigid boundary condition influence particle arrangements at the edges especially when adopting a non-zero interparticle friction coefficient and zero wall-particle friction coefficient. It can also be observed that the distribution tends to be more uniform with increasing sample size for both boundary conditions, while the change is negligible with periodic boundary when sample size L/d_{max} is larger than 10.

In Fig. 7, the secant modulus ratio is determined as the ratio of the secant compression modulus over the secant compression modulus of the largest considered specimen E_{ref} , for the axial stress range 0–1 MPa or 0–10 MPa of B1 samples. The result indicates that the compression modulus ratio decreases with increasing L/d_{max} and approaches a constant when L/d_{max} exceeds 20 under the rigid boundary condition, whereas the compression modulus is almost the same with a L/d_{max} larger than 5 under the periodic boundary. The influence of sample size on the compression modulus is more pronounced for loose specimens at lower axial stress.

Previous studies have employed various factors to understand the size effect including relative density (Varadarajan et al., 2003; Cantor and Ovalle, 2023; He et al., 2023) and void ratio (Kuhn and Bagi, 2009; Alonso et al., 2012; Frossard et al., 2012; Huang et al., 2014). Fig. 8 shows the variation of the secant compression modulus ratio with varying L/d_{max} for specimens of similar relative densities for different axial stress ranges of B1 samples. The result indicates that the compression modulus E_s increases as L/d_{max} increases. The normalized compression modulus E_s/E_{ref} approaches monotonically to unity as L/d_{max} increases, and the change is insignificant after L/d_{max} exceeds 20. The result suggests that different factors may lead to inconsistent conclusions on the sample size effect.

To ensure the repeatability of generated specimens, four different initial packings with a similar void ratio were generated for each size and boundary condition. Fig. 9 shows the variation of secant compression modulus E_s for different initial packings for two axial stress ranges 0–1 MPa and 0–10 MPa, respectively. For the rigid boundary condition, as the L/d_{max} increases, the difference between the compression moduli of different initial packings decreases and vanishes when L/d_{max} reaches 15. The result suggests that the influence of packing on the mechanical behavior of particle assemblies continues to decrease as the sample size increases. However, the different initial packings do not show significant influence on the mechanical behavior of particle assemblies under the periodic boundary condition. Additionally, the secant compression modulus shows less variability at higher axial stress levels compared to those at lower axial stress levels.

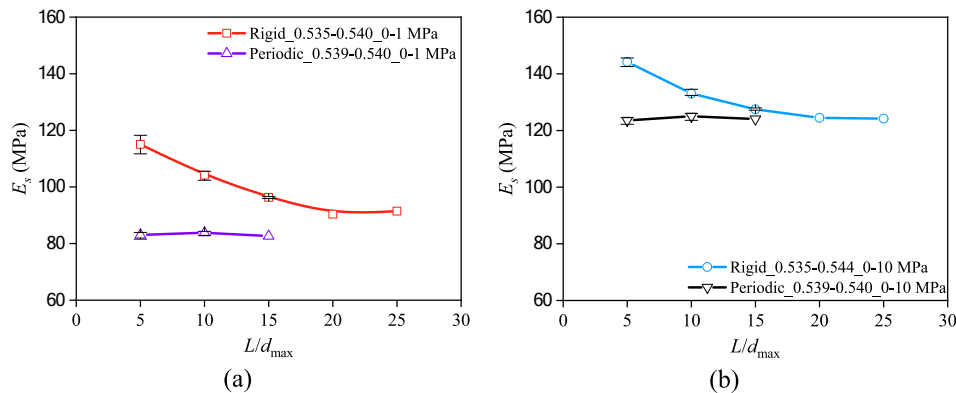


Fig. 9. Variation of compression modulus at axial stress of (a) 0–1 MPa and (b) 0–10 MPa with different sample sizes for B1 samples.

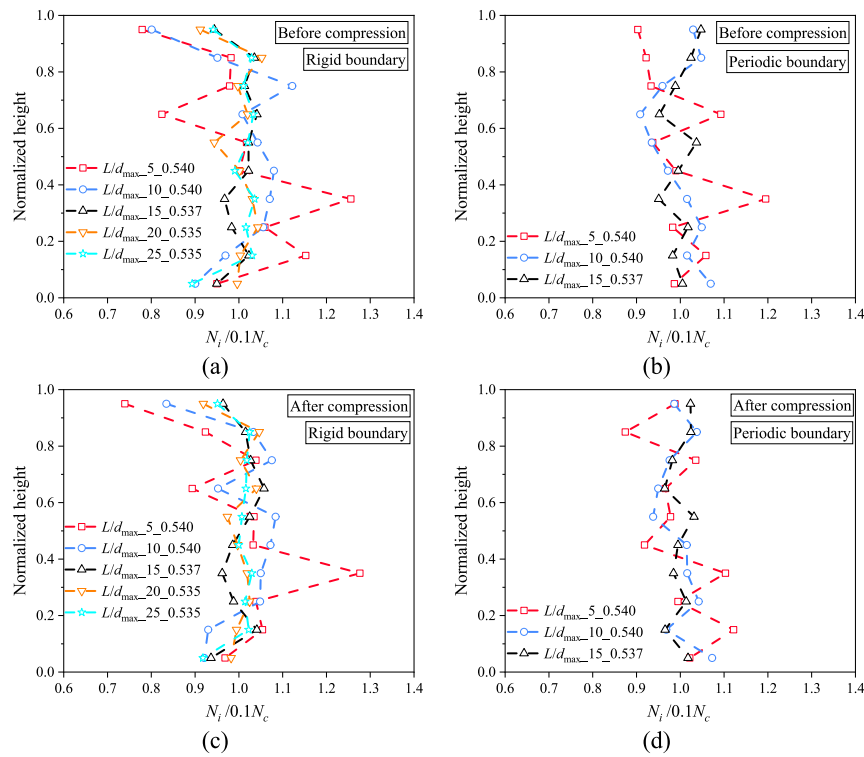


Fig. 10. Comparison of contact number distributions in the vertical direction at different sample sizes L/d_{max} for B1 samples under (a) rigid and (b) periodic boundary conditions before compression, and (c) rigid and (d) periodic boundary conditions after compression.

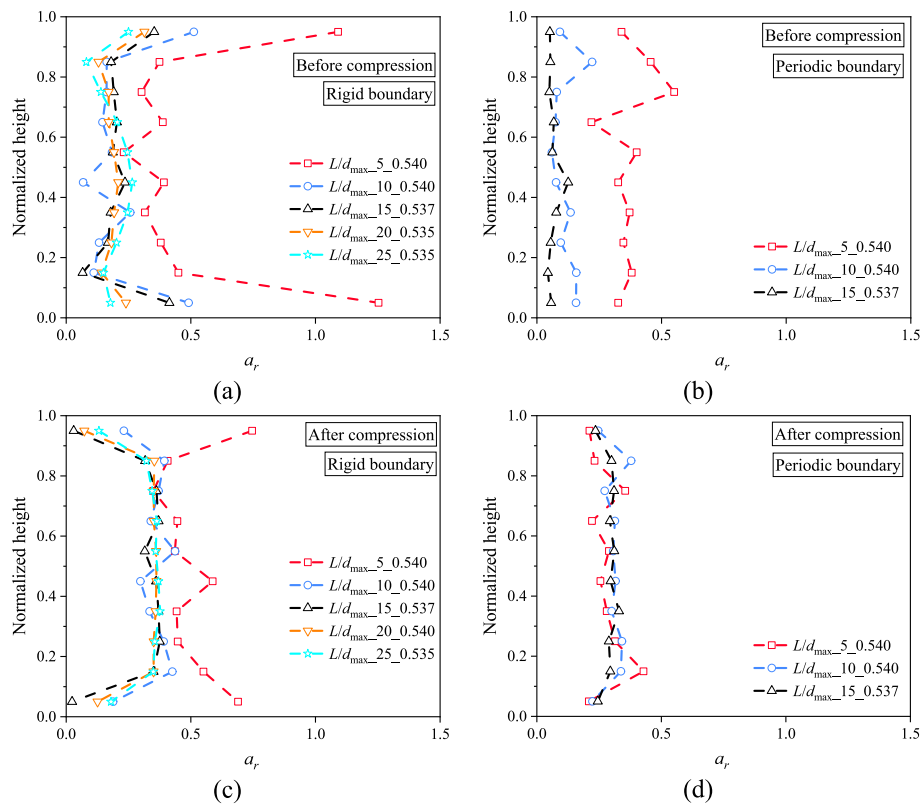


Fig. 11. Distribution of fabric anisotropy factor a_r in the vertical direction for different L/d_{max} for B1 samples under (a) rigid and (b) periodic boundary conditions before compression, and (c) rigid and (d) periodic boundary conditions after compression.

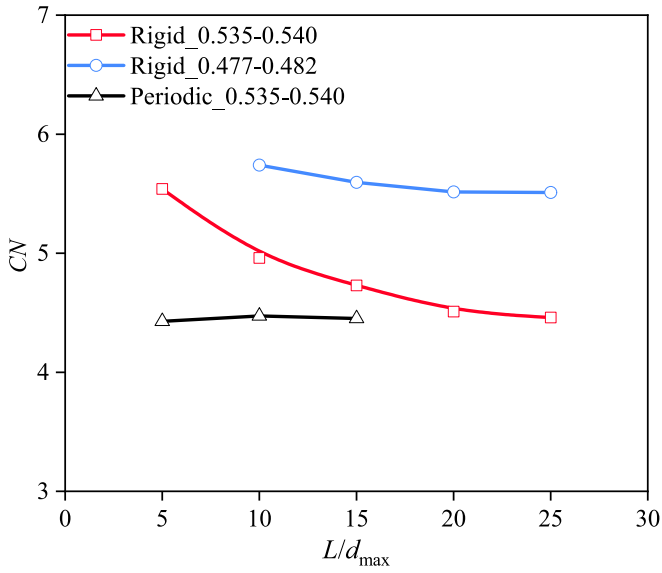


Fig. 12. Variation of coordination number before compression for different $L/d_{\max}$ at different void ratios and two boundary conditions for B1 samples.

3.2. Microscopic interpretation

The microscopic investigation of the size effect is conducted by studying the distribution of contact number and fabric anisotropy factor. Fig. 10 shows the comparison of contact number distribution in the

vertical direction with different sample sizes $L/d_{\max}$ under both rigid and periodic boundaries before and after compression for B1 particle samples. Note that the computation of microscopic quantities was carried out for ten layers of equal height. N_i is the number of all contacts between particles in the i th layer and N_c is the total number of contacts. The result shows that contact distribution becomes more uniform as sample size increases. Meanwhile, the distribution of the sample under periodic boundary condition is more uniform than that under rigid boundary condition at the same sample size $L/d_{\max}$. Additionally, the result shows axial stress has no significant influence on the distribution of contacts.

The commonly adopted fabric tensor quantifying the distribution of the contact normal (Rothenburg and Bathurst, 1992; Gu et al., 2015) is

$$R_{ij} = \int_{\Omega} E(\Omega) n_i n_j d\Omega = \frac{1}{N_c} \sum_{c \in N_c} n_i n_j \quad (1)$$

where n_i is the i th component of the contact normal, N_c is the total number of contacts, and $E(\Omega)$ is the distribution function on the unit sphere Ω whose first-order approximation is

$$E(\Omega) = \frac{1}{4\pi} [1 + \alpha_{ij}^r n_i n_j] \quad (2)$$

where α_{ij}^r is the second-order anisotropy tensor quantifying the degree of fabric anisotropy. α_{ij}^r is related to the deviatoric part of the fabric tensor R'_{ij} as

$$\alpha_{ij}^r = \frac{15}{2} R'_{ij} \quad (3)$$

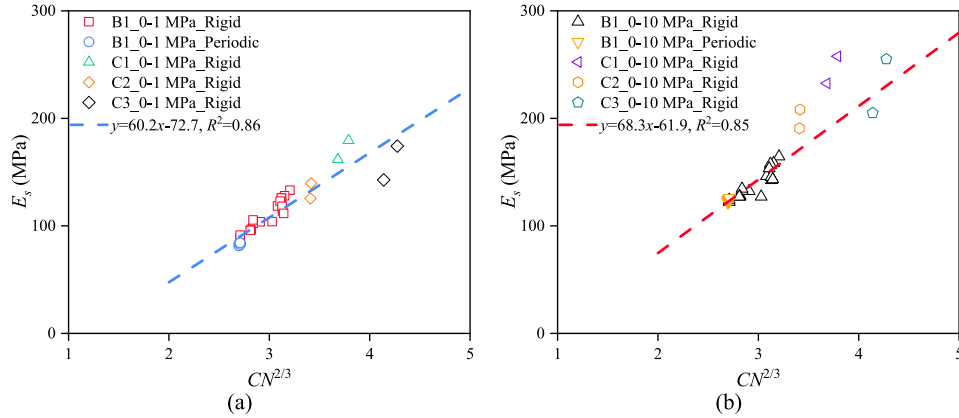


Fig. 13. Correlation between coordination number $CN^{2/3}$ and secant compression modulus E_s at axial stress of (a) 0–1 MPa and (b) 0–10 MPa.

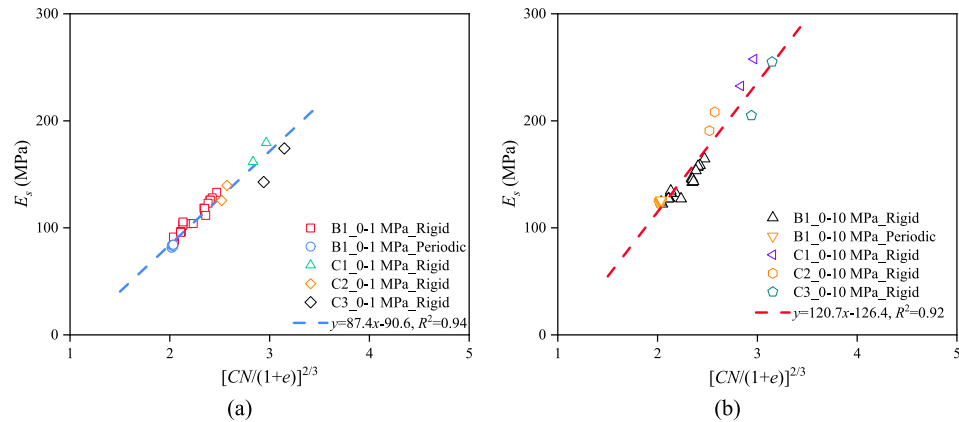


Fig. 14. Correlation between $[CN/(1+e)]^{2/3}$ and secant compression modulus E_s at axial stress of (a) 0–1 MPa and (b) 0–10 MPa.

The scalar parameter α_r is adopted in the following to quantify the degree of fabric anisotropy

$$\alpha_r = \sqrt{\frac{3}{2} \alpha_{ij}^r \alpha_{ij}^r} \quad (4)$$

Fig. 11 compares the distribution of fabric anisotropy factor α_r in the vertical direction with different sample sizes $L/d_{\max}$ under rigid and periodic boundary conditions before and after compression. The results show that the sample becomes more uniform in terms of the contact normal anisotropy as the sample size increases. The fabric anisotropy factor α_r of the top and bottom layers decreases during compression for the sample under rigid boundary condition, and the distribution of fabric anisotropy factor α_r is more uniform after compression. Besides, the fabric anisotropy factor α_r for the sample under periodic boundary condition is more uniform than that under rigid boundary condition.

The mechanical coordination number CN is a significant microscopic parameter, which is defined as

$$CN = \frac{2(C - N_1)}{N - N_1 - N_0} \quad (5)$$

where C is the number of contacts, N is the number of particles, N_1 is the number of particles with only one contact, and N_0 is the number of particles without any contact.

Fig. 12 shows the variation of coordination number before compression for different $L/d_{\max}$ under different void ratios and two boundary conditions for B1 samples. With an increasing sample size, the coordination number decreases under rigid boundary, while that remains almost the same under periodic boundary. Moreover, dense specimens exhibit a smaller reduction in coordination number compared to loose specimens. The evolution of the coordination number agrees with the macroscopic mechanical behavior as shown in Fig. 7. To further validate the result, Fig. 13 presents the correlation between secant compression modulus E_s and the initial coordination number CN for different particle shapes. The result shows that the secant compression modulus E_s increases with the increasing coordination number CN for all the considered sample sizes. Moreover, $CN^{2/3}$ is adopted to explore the linear relationship with E_s (Gu and Yang, 2013). The result suggests that variation in the coordination number could be considered as the primary microscopic measure of the effect of the boundary conditions on the mechanical characteristics of the particulate assemblies. To further validate the results, the coordination number per unit volume $CN/(1 + e)$ proposed by Chang et al. (2005) is introduced in Fig. 14. A better relationship can be observed by considering the void ratio e , which indicates the coordination number per unit volume $CN/(1 + e)$ can serve as a microscopic caliber of the particulate size effect.

4. Conclusions

In this study, various sample sizes, particle shapes, void ratios, and boundary conditions were adopted in the DEM simulations to investigate the size effect on the compressibility of rockfill materials. The microscopic quantities including contact number and fabric anisotropy factor were studied. The major findings are summarized as follows:

- (1) The compressibility increases with increasing $L/d_{\max}$ for rigid boundary condition until $L/d_{\max}$ reaches 20 after which compressibility does not change significantly. Besides, the sample size shows a more significant influence on looser samples. However, the sample size has a negligible influence on the compressibility of rockfill materials with periodic boundary condition.
- (2) The difference between the compression modulus for different initial packings decreases until $L/d_{\max}$ reaches 15, which suggests that different initial packings do not show a significant impact on

the mechanical properties of a large sample of rockfill materials. Besides, initial packing does not show a significant influence on the mechanical properties of rockfill materials with periodic boundary condition.

- (3) Both the contact number and fabric anisotropy factor become more uniform as the sample size increases. Specimens under periodic boundary exhibit greater uniformity. Furthermore, a linear correlation between the compression modulus and coordination number per unit volume $CN/(1 + e)$ can be observed for the considered stress range for the concerned sample sizes and particle shape, which could potentially serve as a microscopic caliber of the particulate size effect.

CRediT authorship contribution statement

Hongwei Wu: Writing – original draft, Investigation. **Xiaoqiang Gu:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Jing Hu:** Writing – review & editing, Validation, Methodology, Funding acquisition, Conceptualization. **Gang Deng:** Writing – review & editing. **Fang Liu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The work is supported by National Natural Science Foundation of China (Grant no. 52208364), IWH Research and Development Support Program (Grant no. GE0145B032021) and Shanghai Pujiang Program (Grant no. 22PJJD074). These supports are gratefully acknowledged.

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